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Question Paper Code : 17991

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2021

Third Semester

B.E. – Computer Science and Engineering

19CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

BTL, CO		
U, CO1	1.	Consider a hypothetical processor with an instruction of type LW R1, 20(R2), which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for operand in memory? Justify your answer. A) Immediate Addressing B) Register Addressing C) Register Indirect Scaled Addressing D) Base Indexed Addressing
R, CO1	2.	List out different types of addressing modes.
U, CO2	3.	Consider Z = X – Y, where X, Y and Z are all in sign-magnitude form. X and Y are each represented in *n* bits. To avoid overflow, the representation of Z would require a minimum of _____. Justify your answer. A) n bits B) n-1 bits C) n+1 bits D) n+2 bits
R, CO2	4.	List various fast multiplication methods.
A, CO3	5.	Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I1, I2, I3, …, I12 is executed in this pipelined processor. Instruction I4 is the only branch instruction and its branch target is I9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is _____. Justify your answer. A) 132 B) 165 C) 176 D) 328

A, CO3 6. Say true/false, there is a data dependency between the following instructions: Justify your answer.

Store R1, 100(R2)

Load R1, 110(R2)

U, CO4 7. In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from _____. Justify your answer

A) (j mod v) * k to (j mod v) * k + (k-1)

B) (j mod v) to (j mod v) + (k-1)

C) (j mod k) to (j mod k) + (v-1)

D) (j mod k) * v to (j mod k) * v + (v-1)

R, CO4 8. By what ways, Static RAM and Dynamic RAM differ?

U, CO5 9. Why IO devices cannot be directly connected to the system bus?

R, CO5 10. Mention various interconnection standards.

PART B – (5 × 16 = 80 Marks)

U, CO1 11. a) i. Summarize the basic functional units of a computer. (8 Marks)

A, CO1 11. a) ii. List the steps needed to execute the machine instruction Add LOCA, R0 in terms of transfer between processor and the memory along with some simple control commands.

Assume that the instruction itself is stored in the memory at location INSTR and that this address initially in registers PC. The first two steps might be expressed as:

A) Transfer the content of register PC to register MAR

B) Issue read command to the memory and then waits until it has transferred the requested word int register MDR. (8 Marks)

OR

U, CO1 11. b) i. Categorize and describe the types of addressing modes with examples. (8 Marks)

A, CO1 11. b) ii. Write an assembly language program to add a scalar value to all the elements of a collection of values. (8 Marks)

U, CO2	12. a) i.	Elaborate how to multiply two unsigned binary numbers with an example.	(8 Marks)
A, CO2	12. a) ii.	Multiply the following using bit-pair recording method: (-12) x 5.	(8 Marks)
		OR	
U, CO2	12. b) i.	Explain the rules for adding, subtracting, multiplying and dividing two floating point numbers with examples.	(8 Marks)
A, CO2	12. b) ii.	Divide the following using Non-restoring integer division algorithm: 12/5	(8 Marks)
U, CO3	13. a) i.	Explain the working of Hardwired control.	(8 Marks)
U, CO3	13. a) ii.	How is the performance of pipeline improved, when there is a data dependency? Explain the methods.	(8 Marks)
		OR	
U, CO3	13. b) i.	Discuss 3 bus architecture of CISC style processor with a neat diagram.	(8 Marks)
U, CO3	13. b) ii.	Determine the steps involved in fetching and executing the instruction MUL R1, R2, R3.	(8 Marks)
U, CO4	14. a) i.	Elaborate on direct memory access.	(8 Marks)
U, CO4	14. a) ii.	Propose the types of Read-Only Memory and explain each.	(8 Marks)
		OR	
U, CO4	14. b) i.	Discuss the different mapping techniques used in cache memories and their relative merits and demerits.	(8 Marks)
U, CO4	14. b) ii.	Examine the organization of magnetic disk in detail.	(8 Marks)
U, CO5	15. a) i.	Explain in detail about bus arbitration with a neat diagram	(8 Marks)
U, CO5	15. a) ii.	With a neat diagram, explain serial interface in interface circuits.	(8 Marks)
		OR	
U, CO5	15. b) i.	Discuss in detail about interrupts.	(8 Marks)
U, CO5	15. b) ii.	Outline the working principle of USB.	(8 Marks)